

Reviewer Pack — Gowers Uniformity Norm of Boolean Functions Bounds ACC⁰ Circ...

Entry b61cbcb3e1c4 · INCONCLUSIVE

SEC autonomous research engine — attribution: Ludovico Kubler

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Gowers Uniformity Norm of Boolean Functions Bounds ACC⁰ Circuit Size

Entry ID: b61cbcb3e1c4 **Verdict:** INCONCLUSIVE **Recorded:** 2026-05-08 07:21:55 UTC

1. Conjecture

Field A (mathematical branch): Gowers Uniformity Norms in Additive Combinatorics **Field B** (complexity object): ACC⁰ Circuit Size

Statement:

For a Boolean function f on n variables, if the Gowers uniformity norm of order 3 is at least $\Omega(n^\epsilon)$, then the minimal ACC⁰ circuit size for f is $\Omega(n^{\{2-\epsilon\}})$

Rationale (proposer's reasoning):

High Gowers uniformity indicates the function is not well-approximated by low-degree polynomials, which are efficiently computable by ACC⁰ circuits. Thus, such functions require larger ACC⁰ circuits.

Taxonomy category: ACC_SIPSER (status at proposal time: alive)

2. Pre-registration (Popper-style)

Hash (SHA-256 prefix): 9f9a3b88e124e9f3

This hash commits to the conjecture statement, acceptance criterion, and seed list **before** the empirical test runs. Tampering with the test or its data after this hash was computed would be detectable.

Acceptance criterion:

default: $\geq 80\%$ seeds must support, no counterexample

3. Barrier filter (F1)

Final: PASS

Barrier	Verdict	Confidence	LLM ₁	LLM ₂
RELATIVIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
NATURAL_PROOFS	SAFE	0.00	UNCERTAIN	UNCERTAIN
ALGEBRIZATION	SAFE	0.00	UNCERTAIN	UNCERTAIN
KARP_LIPTON	SAFE	0.00	UNCERTAIN	UNCERTAIN

4. Novelty audit

Verdict: NOVEL against 0 hits across arXiv + Semantic Scholar + ECCC

5. Empirical test harness

Multi-seed protocol: 5 seeds (11, 23, 37, 53, 71)

Sandbox: isolated subprocess, stdlib-only, ≤ 90 s wall time per cycle **Execution:** rc=124, elapsed=240.0s

5.1 Generated Python source

```
import random
import math
from itertools import product

def fourier_coefficients(f, n):
    N = 2 ** n
    coeffs = [0] * N
    for k in range(N):
        sum_val = 0
        for x in range(N):
            sum_val += f(x) * math.cos(2 * math.pi * k * x / N)
        coeffs[k] = sum_val / N
    return coeffs

def gowers_uniformity_norm(f, n):
    coeffs = fourier_coefficients(f, n)
    norm = 0
    for coeff in coeffs:
        norm += abs(coeff) ** 4
    return norm ** (1/4)

def simulate_acc0_circuit(f, n, size):
    # Simplified simulation of ACC^0 circuit with given size
    # This is a placeholder and does not actually compute the function
    return False

def run_trial(seed: int) -> dict:
    random.seed(seed)
    n = random.choice([5, 10, 15, 20, 30, 40])
    instances_tested = 0
    conjecture_holds = True
    counterexample = ""

    for _ in range(50): # Sample 50 instances per seed
        f = lambda x: random.choice([0, 1]) # Random Boolean function
        norm = gowers_uniformity_norm(f, n)
        if norm >= n ** 0.1: # Threshold for  $\epsilon=0.1$ 
            circuit_size = simulate_acc0_circuit(f, n, size=n**2)
            if not circuit_size:
                conjecture_holds = False
                counterexample = "Function cannot be computed by ACC^0 circuit"
                break

    instances_tested += 1
```

```

    return {
        "metric_name": "Gowers Uniformity Norm",
        "metric_value": norm,
        "instances_tested": instances_tested,
        "conjecture_holds": conjecture_holds,
        "counterexample": counterexample
    }

if __name__ == "__main__":
    import sys
    seeds = [int(s) for s in sys.argv[1:]] or [2**i - 1 for i in range(5, 8)] # Default to first

    results = []
    for seed in seeds:
        result = run_trial(seed)
        print(f"TRIAL: {{\"seed\": {seed}, **{result}}}")
        results.append(result)

    mean_value = sum(r["metric_value"] for r in results) / len(results)
    std_value = math.sqrt(sum((r["metric_value"] - mean_value) ** 2 for r in results) / len(results))
    support_fraction = sum(1 for r in results if r["conjecture_holds"]) / len(results)

    if all(r["conjecture_holds"] for r in results):
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    elif support_fraction >= 0.8:
        print(f"RESULT: SUPPORTED mean={mean_value} std={std_value} support_fraction={support_fraction}")
    else:
        first_failing_seed = next((r["seed"] for r in results if not r["conjecture_holds"]), None)
        print(f"RESULT: FALSIFIED counterexample=\"{counterexample}\" first_failing_seed={first_failing_seed}")

```

6. Per-seed results

(no seed-level data — test crashed before producing TRIAL: lines)

7. Test stdout (last 2KB)

TIMEOUT after 240s

8. Critic adversarial review

Critic verdict: CHALLENGE

Critic reasoning:

skipped: test crashed before producing data

9. Final verdict & safety rail

Verdict: INCONCLUSIVE

Reasoning:

Test timed out before producing results, preventing evaluation of support fraction or counterexamples. | next: Increase timeout duration and re-run test with identical parameters

11. Audit log (LLM calls)

Total LLM calls: 7

#	Phase	Provider	Model	Tokens in	out	Latency (ms)
1	propose	ollama_remote	qwen3:8b	0	0	72907
2	preregistration	ollama_remote	qwen3:8b	0	0	20002
3	novelty	ollama_remote	qwen3:8b	0	0	16571
4	novelty	ollama_remote	qwen3:8b	0	0	8603
5	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	15218
6	test_gen	ollama_remote	qwen2.5-coder:7b	0	0	8575
7	judge	ollama_remote	qwen3:8b	0	0	11887

Totals: 0 input tokens, 0 output tokens, ~\$0.0000 cost, 153763 ms total latency. Provider mix: {'ollama_remote': 7}

(full prompt+response transcripts available in `research/audit/b61cbcb3e1c4.jsonl`)

12. Reproducibility

To reproduce this cycle:

1. Apply the test harness in section 5.1 with seeds 11,23,37,53,71
2. The Python sandbox is pure-stdlib; any Python 3.11+ should suffice
3. The full LLM transcripts are in `research/audit/b61cbcb3e1c4.jsonl` for verification of provenance
4. The pre-registration hash in section 2 commits to the test design before execution; tampering would be detectable.

Sandbox file: `research/pvsnp_sandbox/test_*.py` (per-cycle)

Replay tarball: `research/replay/b61cbcb3e1c4.tar.gz` (if generated)